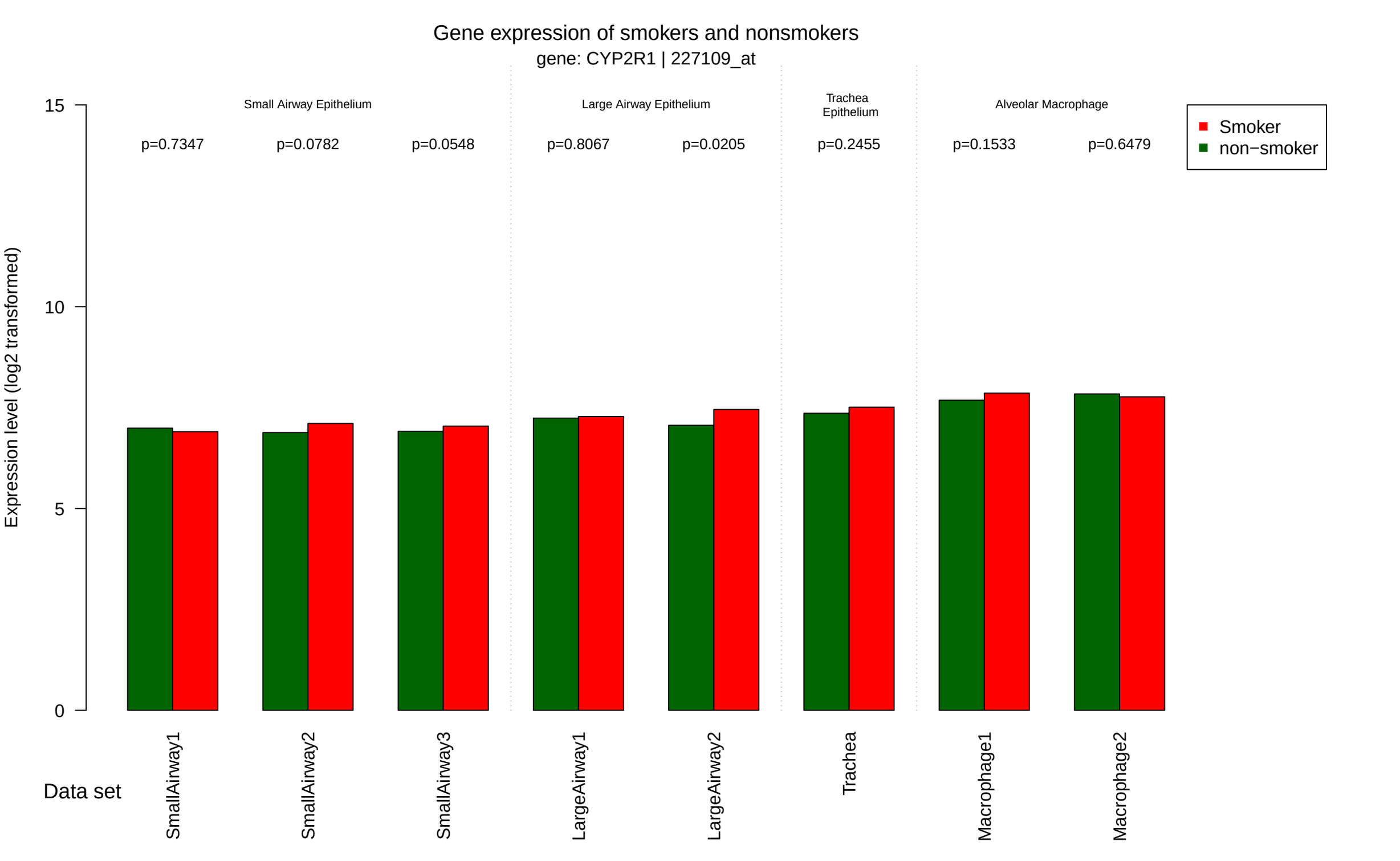
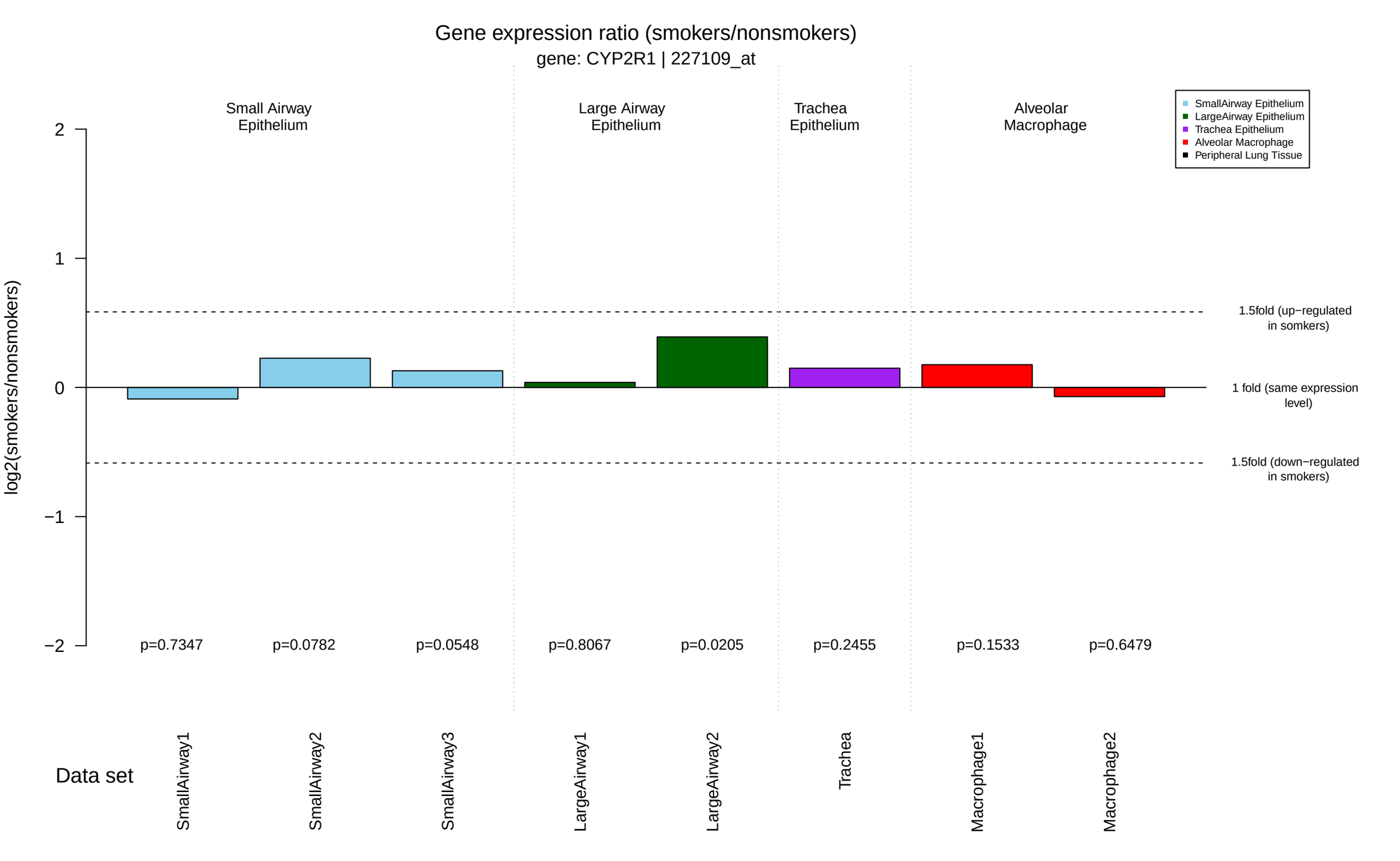
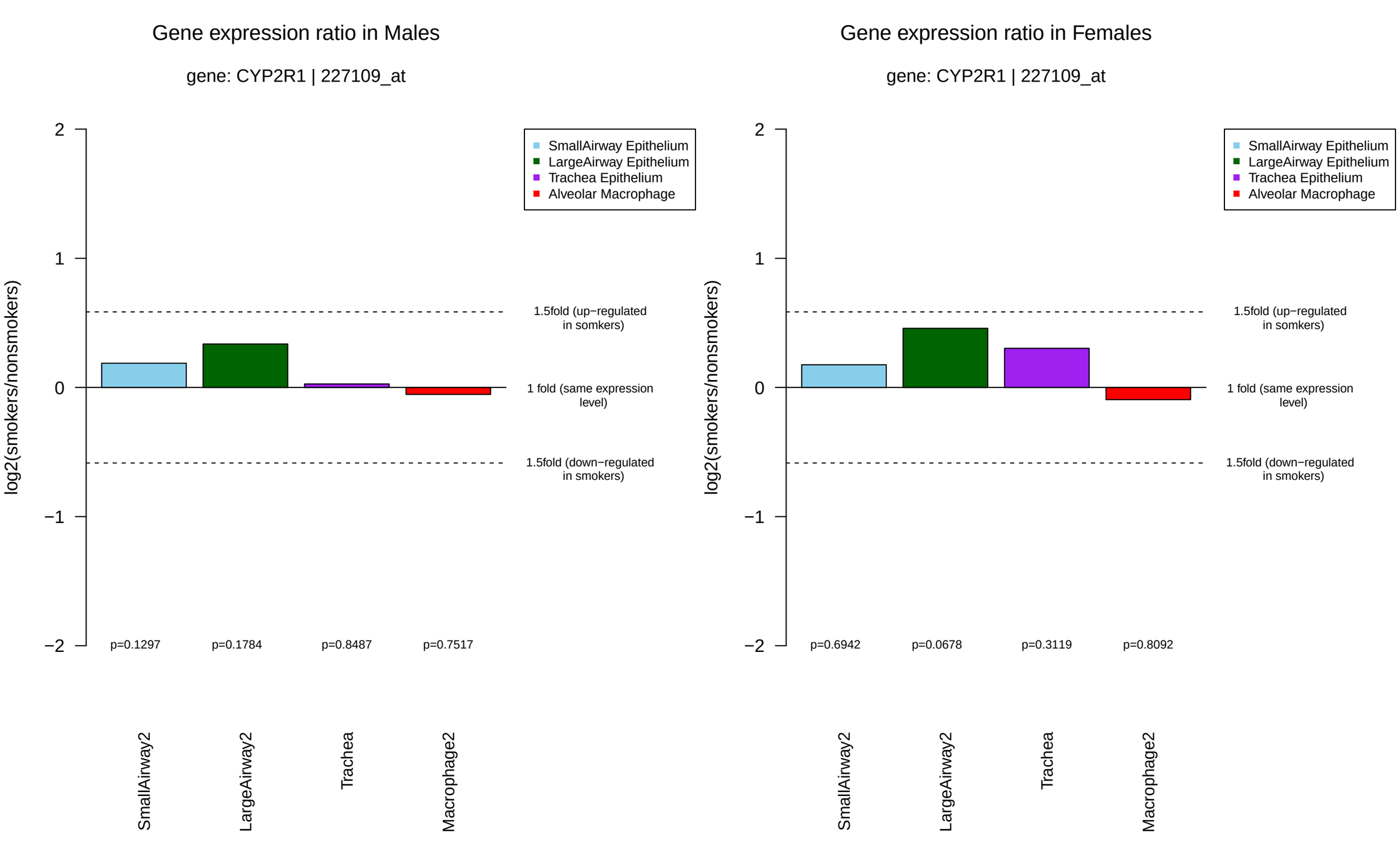
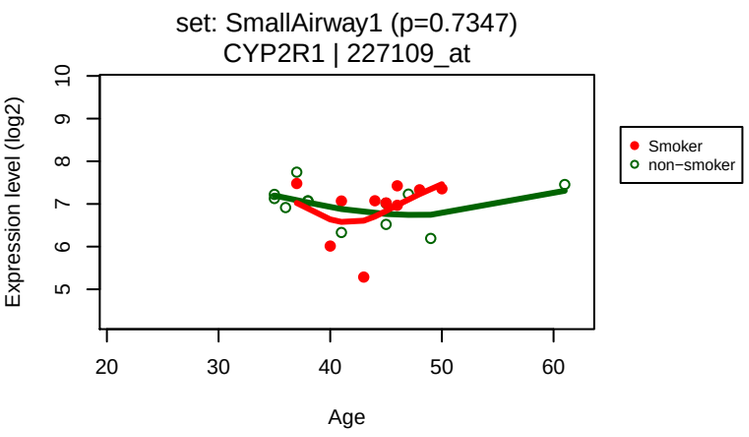
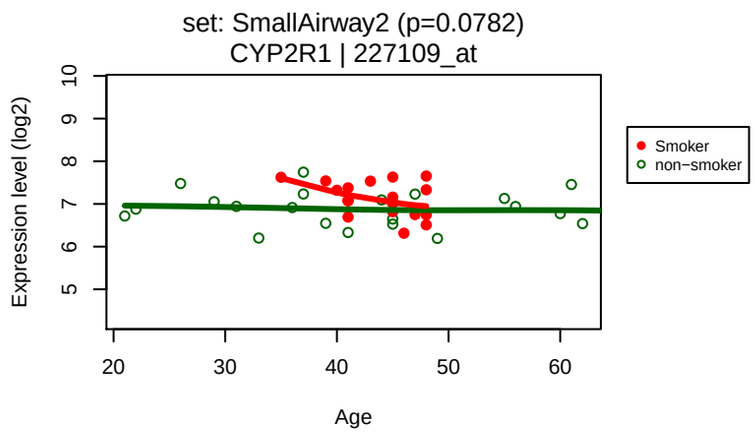
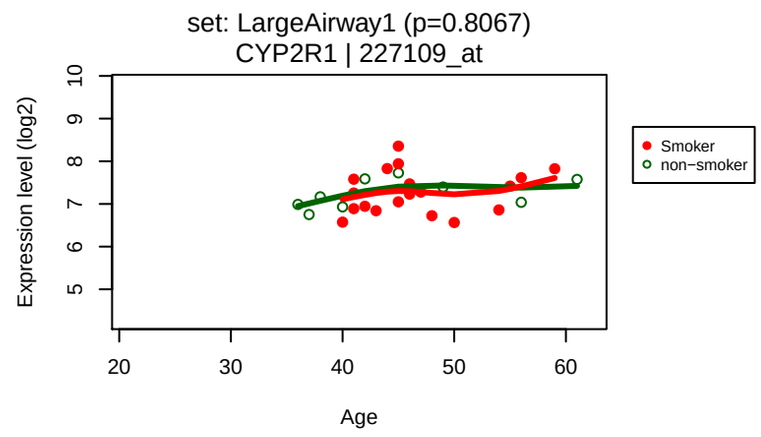

Gene expression of smokers and nonsmokers
gene: CYP2R1 | 227109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7347
p=0.0782
p=0.0548
p=0.8067
p=0.0205
p=0.2455
p=0.1533
p=0.6479
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

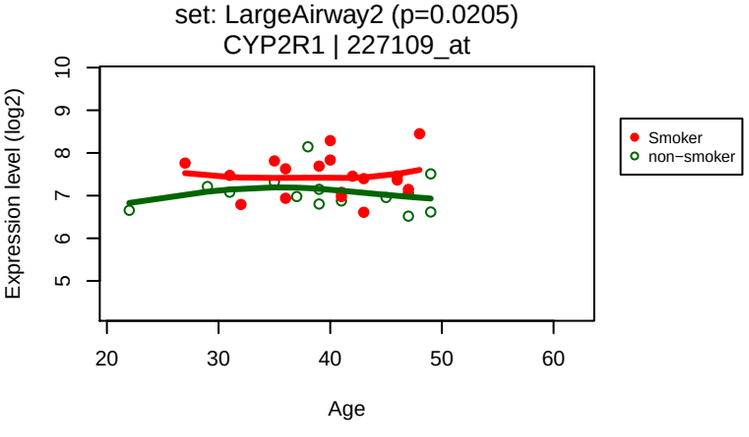

Gene expression ratio (smokers/nonsmokers)
gene: CYP2R1 | 227109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7347
p=0.0782
p=0.0548
p=0.8067
p=0.0205
p=0.2455
p=0.1533
p=0.6479
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

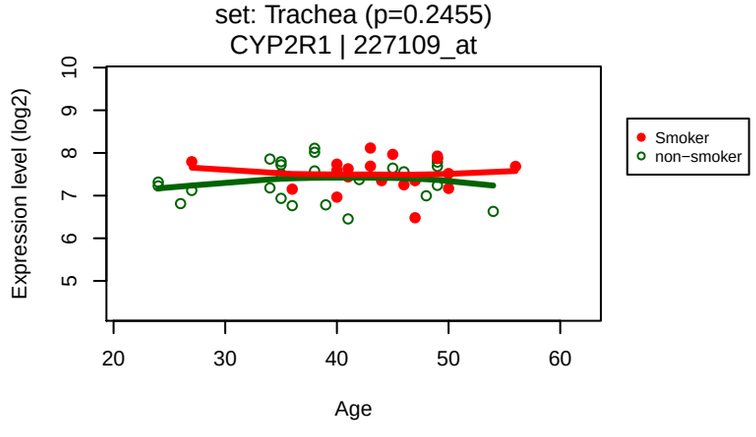

Gene expression ratio in Males
gene: CYP2R1 | 227109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1297
p=0.1784
p=0.8487
p=0.7517
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2R1 | 227109_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6942
p=0.0678
p=0.3119
p=0.8092
SmallAirway2
LargeAirway2
Trachea
Macrophage2

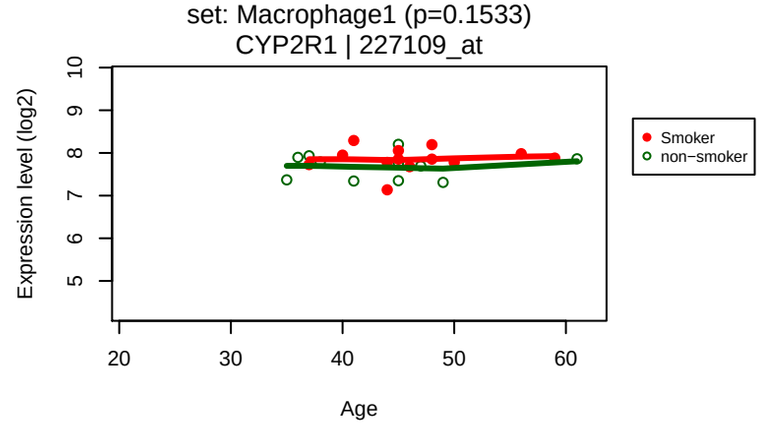

set: SmallAirway1 (p=0.7347)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker

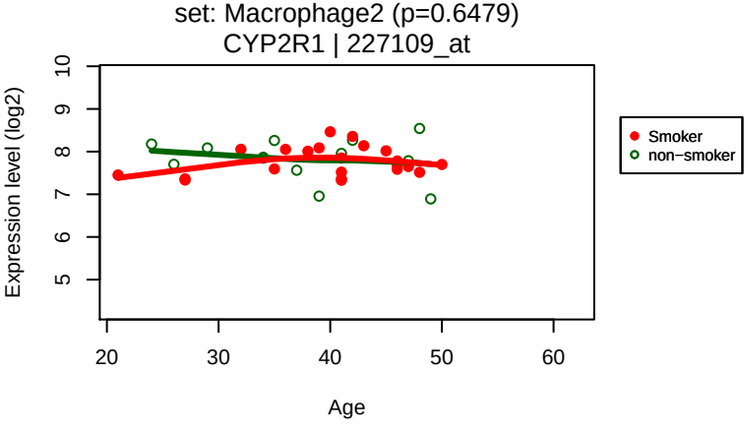

set: SmallAirway2 (p=0.0782)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker

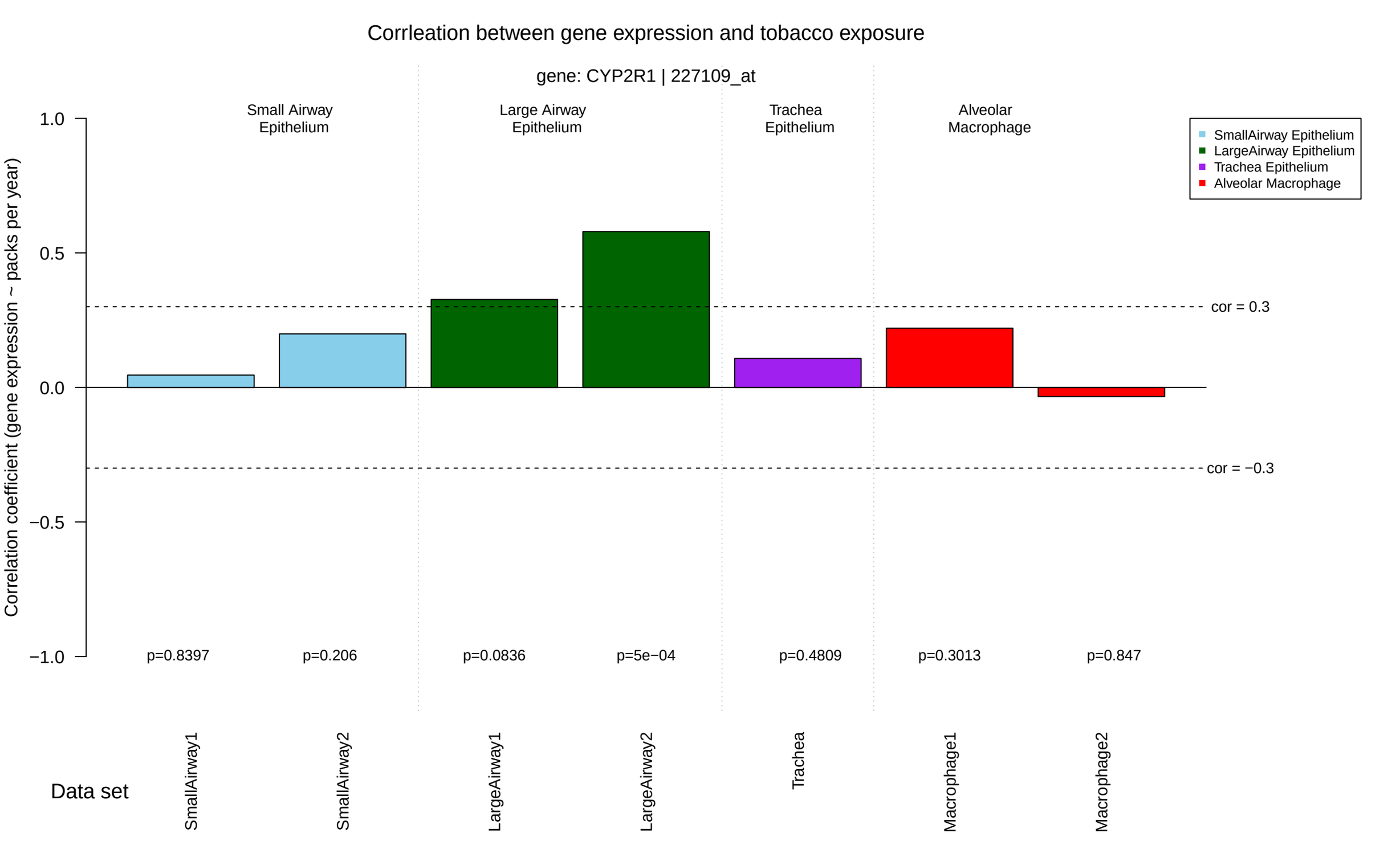

set: LargeAirway1 (p=0.8067)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0205)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2455)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1533)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6479)
CYP2R1 | 227109_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2R1 | 227109_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8397
p=0.206
p=0.0836
p=5e−04
p=0.4809
p=0.3013
p=0.847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2